

Draft "Evaluation" section (paper-ready outline)

Purpose: a structured, factual scaffold an LLM/author only has to rephrase and condense into the paper.

All numbers are final and reproducible. Keep the *evaluated vs mentioned* framing - it is a selling point.

X. Evaluation of Extraction Quality

X.1 Setup

We evaluate the automatically populated NS4KGE knowledge graph against the source articles, for the four entity types it contains: **datasets, metrics, KGE models, and NS methods**, over a corpus of **55 articles**.

Crucially, the evaluation is **faithful to the extraction pipeline**: each entity is checked against the exact input its generating prompt received. The pipeline splits every article into a *tables-only* view (`load_md_tables_only`, feeding the results-table prompt) and a *no-tables* view (`load_md_no_tables`, feeding the metadata prompt). Datasets and metrics originate solely from results tables, so they are validated against the tables-only view; KGE models and NS methods originate from **both** the tables (configurations = evaluated entities) and the prose (proposed/mentioned entities), and are validated against the corresponding view. These views are regenerated with the pipeline's own functions, guaranteeing no train/eval mismatch.

The evaluation procedure is **fully deterministic** (regular-expression matching, no LLM), hence exactly reproducible.

X.2 Metrics

For each entity type we report:

- **Precision** = $TP / (TP + FP)$: fraction of extracted entities actually attested in the source.
- **Relative recall** = $TP / (TP + FN \text{ candidates})$: fraction of attested entities that were extracted, where candidates are drawn from the global vocabulary of entities extracted across the corpus. Recall is *relative* (a lower-bound proxy): candidates are text-attested entities that were not extracted.

For KGE models and NS methods we further split every metric by **provenance** - *evaluated* (appearing in a results table) vs *mentioned only* (cited in prose) - yielding four scores. This mirrors an ontological distinction the KG is meant to capture: which methods/models are experimentally compared vs merely discussed.

X.3 Matching

Matching is case-insensitive with tolerant separators and strict word boundaries (a name may not match a prefix of a longer code, e.g. FB15k vs FB15k-237). Difficulty grows with the lack of name standardization, requiring type-specific handling:

- **Datasets**: verbatim; recall restricted to table cells (results vs statistics tables separated by caption).
- **Metrics**: deterministic alias set, since the prompt normalizes surface forms (Mean Rank->MR, Hit@10->Hits@10).
- **KGE models**: case-sensitive on the recall side (names are stylized as words, e.g. Simple); generic architectures (GAN, MLP, BERT) excluded as non-instances of the class.
- **NS methods**: acronym derivation (Self-Adversarial Negative Sampling->SANS), qualifier normalization (NoiGAN (hard) (40%)->NoiGAN), near-duplicate canonicalization, and the placeholder Unknown excluded.

X.4 Results

Entity	Prec. evaluated	Prec. mentioned	Recall evaluated	Recall mentioned
Datasets	98.9%	-	97.3%	-
Metrics	96.5%	-	96.0%	-
KGE Models	94.5%	97.9%	89.5%	95.5%
NS Methods	89.0%	87.0%	80.9%	71.2%

Precision is high across all four types ($\geq 87\%$), and the few precision errors are genuine extraction defects that the evaluation localizes (e.g. a hallucinated Accuracy metric, unsplit model+method compounds).

X.5 Analysis / Discussion

The scores decrease monotonically with the **degree of standardization of entity names**. Dataset and KGE-model

names are frozen community nomenclature (WN18RR, TransE), matched near-perfectly. Metrics form a small closed set, only lightly rewritten. NS methods, by contrast, lack a shared naming convention: many are paraphrased or have no canonical short name, are heavily abbreviated, and are frequently confounded with inferred, unnamed baselines - which explains, and largely justifies, their lower scores. The lower NS-method recall on mentions further quantifies a concrete, actionable behavior of the KG: it misses about one third of the methods cited in related work.

X.6 Limitations

Recall is *relative* (evaluated against the extracted global vocabulary, not an exhaustive gold standard).

Precision is reported strictly: inferred standard baselines that the article does not name explicitly are counted as errors even though they may be correct, since they cannot be confirmed by textual search - a limitation specific to NS methods.